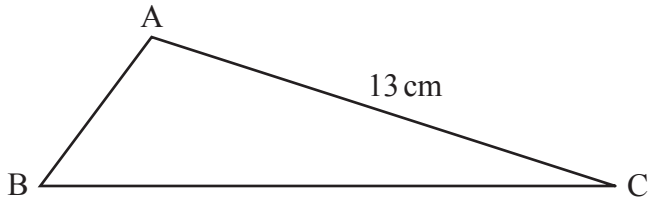
5 The diagram shows the triangle ABC in which $AC = 13\,\text{cm}$ and AB is the shortest side. The perimeter of the triangle is 32 cm. The area is $24\,\text{cm}^2$ and $\sin B = \frac{4}{5}$.



Determine the lengths of AB and BC. **[5]**